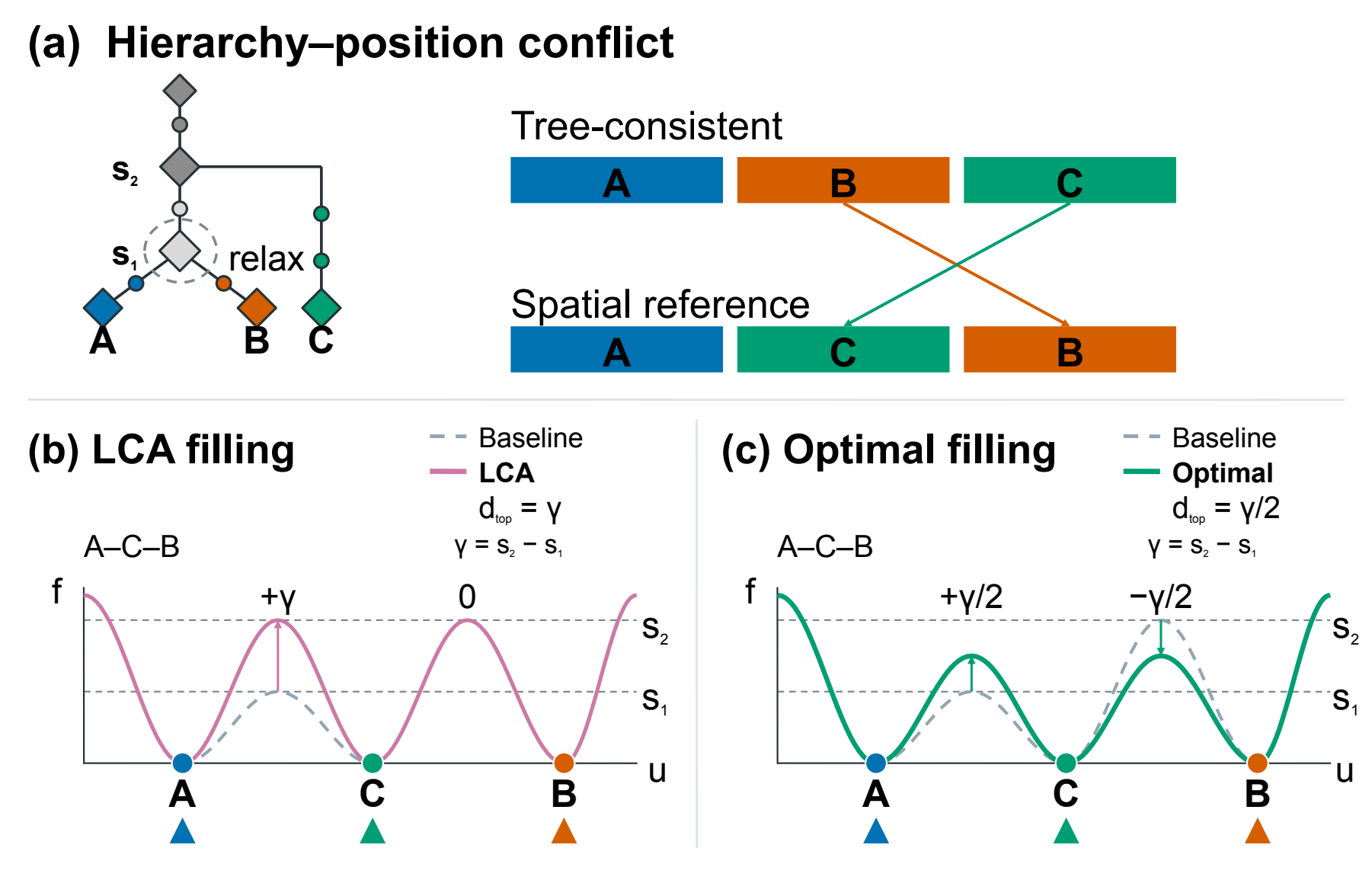

(a) Hierarchy–position conflict
s2
s1
relax
A
B
C
Tree-consistent
A
B
C
Spatial reference
A
C
B
(b) LCA filling
Baseline
LCA
dtop = γ
γ = s2 − s1
A–C–B
f
+γ
0
s2
s1
u
A
C
B
(c) Optimal filling
Baseline
Optimal
dtop = γ/2
γ = s2 − s1
A–C–B
f
+γ/2
−γ/2
s2
s1
u
A
C
B